Origins of the Universe

THE BIG BANG THEORY EXPLAINS HOW THE UNIVERSE DEVELOPED.

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Although nobody knows how the Universe began, evidence suggests that it started with an ancient burst of energy and is still expanding.

The Big Bang theory

In the 1920s, it was discovered that our galaxy and the millions of galaxies that surround it are all moving away from each other because the Universe is expanding. This implies that the galaxies all began close together, billions of years ago. The idea that the Universe started as a hot burst of energy in the distant past was widely accepted once the remains of that energy were observed in the 1960s. In the 13.7 billion years that have passed since the Universe began, that energy has cooled to –270°C (–454°F), and is now known as cosmic microwave background radiation.

There are two theories that predict the way in which the Universe may end: it might be **torn to pieces** in a "**Big Rip**," or become cold and black in a "**Big Chill**."

▽ **The history of everything**
This diagram shows the story of the Universe. Moving from left to right, the intervals of time become longer: the halfway mark on the picture is 500,000 years, which is only 0.04 percent of the age of the Universe.